

Applied Technology Schools

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First Unit

Calculus

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Second Unit

Statistics

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First Unit

Calculus

Dear student, by the end of studying of this unit you should have the following abilities and knowledges:

- ❖ To determine the method of finding the limit of the function (by using direct substitution, factorization or multiplying by conjugate)
- ❖ To find the limit of the function by using direct substitution.
- ❖ To find the limit of the function by using factorization.
- ❖ To find the limit of the function by using multiplying by conjugate.
- ❖ To find the limit of the function by using the law.
- ❖ To find the limit of the function at infinity.
- ❖ To find the limit of the trigonometrical function.
- ❖ To find the variation function.
- ❖ To find the average rate of change function.
- ❖ To find the rate of change.

Lesson (1)

Pre-test

First Question: Factorize the following:

- (1) $x^2 - 16 = \dots\dots\dots$
- (2) $x^2 - 3x - 10 = \dots\dots\dots$
- (3) $x^3 - 8 = \dots\dots\dots$
- (4) $x^3 + 125 = \dots\dots\dots$
- (5) $x^2 - 10x + 25 = \dots\dots\dots$

Second Question:

- (1) Solve the right-angled triangle at B in which $AB = 6.2$ cm, $m(\angle C) = 52^\circ$
- (2) Find the area of circular segment in a circle of radius length 8 cm and its arc length is 20 cm.

Third Question:

- (1) From the top of a lighthouse of height 50 m, a boat was observed at an angle of depression of measure 35° . Find the distance between the boat and the top of the lighthouse.
- (2) Without using calculator, Find the value of: $\frac{\cos 70^\circ \cos 10^\circ + \sin 70^\circ \sin 10^\circ}{\sin 15^\circ \cos 15^\circ}$

Fourth Question:

- (1) ΔABC in which $a = 6$ cm, $b = 8$ cm and $m(\angle C) = 54^\circ$. Find the perimeter of the triangle to the nearest cm.
- (2) Find the solution set of: $x - y = 6$, $x + y = 2$

Lesson (2)

Revision on what studied before

(1)Factorization by taking the H.C. F.:

The expression = $2x^3 + 4x^2$

(both of 2 and 4 are divisible by 2) (the symbol is taking by smallest power)

The expression = $2x^3 + 4x^2 = 2x^2 (x + 2)$

(2)Factorization of the trinomial:

(a) The expression = $x^2 + 5x + 6$ (the sign of the last term is positive) (we are looking for two numbers, such that their product =6,their sum = 5)

The expression = $(x + 3) (x + 2)$ (the signs are as middle term)

(b) The expression = $x^2 + 5x - 6$ (the sign of the last term is negative) (we are looking for two numbers, such that their product =6,their difference= 5)

The expression = $(x + 3) (x - 2)$

(the signs are different, the of greater number is as middle term)

(3)Factorization of the difference of two squares:

The expression = $x^2 - y^2 = (x - y) (x + y)$

The expression = $x^2 - 4 = (x - 2) (x + 2)$

The expression = $x^2 - 9 = (x - 3) (x + 3)$

(4)Factorization of the difference of two cubes:

The expression $= x^3 - y^3 = (x - y) (x^2 + x y + y^2)$

The expression $= x^3 - 8 = (x - 2) (x^2 + 2x + 4)$

The expression $= x^3 - 27 = (x - 3) (x^2 + 3x + 9)$

(5)Factorization of the sum of two cubes:

The expression $= x^3 + y^3 = (x + y) (x^2 - x y + y^2)$

The expression $= x^3 + 8 = (x + 2) (x^2 - 2x + 4)$

The expression $= x^3 + 27 = (x + 3) (x^2 - 3x + 9)$

(6)Solving two equations of first degree in two variables:

Find the solution set of: $x + y = 2$, $y = x + 2$ in $R \times R$

(a) Substitution method

$$x + (x + 2) = 2$$

$$2x = 0$$

$$x = 0$$

$$y = 0 + 2 = 2$$

$$S.S. = \{(0,2)\}$$

(b) Om

$$x + y = 2$$

$$-x + y = 2$$

$$2y = 4$$

$$y = 2$$

$$x + 2 = 2$$

$$x = 0$$

$$S.S. = \{(0,2)\}$$

Lesson 3

Limit of the function algebraically (using factorization)

To find the limit of the function algebraically:

To find $\lim_{x \rightarrow a} f(x)$  find $f(a)$

If:

(1) $f(a) = L$ then "L" is the limit of the function.

(2) $f(a) = \frac{a}{\text{Zero}}$ then the function has no limit.

(3) $f(a) = \frac{\text{Zero}}{\text{zero}}$ (unspecified quantity) (in this case we have to find limit of the function)

Example 1:

Find: $\lim_{x \rightarrow -2} x^2 - 4$

Solution

$$f(-2) = 4 - 4 = 0$$

$$\therefore \lim_{x \rightarrow -2} x^2 - 4 = 0$$

Example 2:

Find: $\lim_{x \rightarrow 1} \frac{x^2 - 5}{x}$

Solution

$$f(1) = \frac{1 - 5}{1} = -4$$

$$\therefore \lim_{x \rightarrow 1} \frac{x^2 - 5}{x} = -4$$

Example 3:

Find: $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

Solution

$$f(2) = \frac{4 - 4}{2 - 2} = \frac{\text{Zero}}{\text{Zero}}$$

$$\therefore \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x + 2)(x - 2)}{(x - 2)} = \lim_{x \rightarrow 2} x + 2 = 4$$

Example 4:

Find: $\lim_{x \rightarrow 2} \frac{x^2 + 4}{x - 2}$

Solution

$$f(2) = \frac{4 + 4}{2 - 2} = \frac{8}{\text{Zero}}$$

$$\therefore \lim_{x \rightarrow 2} \frac{x^2 + 4}{x - 2} \text{ (has no limit)}$$

Example 5:

Find: $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

Solution

$$f(5) = \frac{25 - 25}{5 - 5} = \frac{\text{Zero}}{\text{Zero}}$$

$$\therefore \lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = \lim_{x \rightarrow 5} \frac{(x - 5)(x + 5)}{(x - 5)}$$

$$= \lim_{x \rightarrow 5} x + 5 = 10$$

Example 6:

Find: $\lim_{x \rightarrow 3} \frac{x^2 - 8x + 15}{x - 3}$

Solution

$$f(3) = \frac{9 - 24 + 15}{3 - 3} = \frac{\text{Zero}}{\text{Zero}}$$

$$\lim_{x \rightarrow 3} \frac{x^2 - 8x + 15}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 5)(x - 3)}{(x - 3)}$$

$$= \lim_{x \rightarrow 3} (x - 5) = -2$$

Example 7:

Find: $\lim_{x \rightarrow 0} \frac{(x+2)^2 - 4}{x^2 + x}$

Solution

$$f(0) = \frac{4 - 4}{0 + 0} = \frac{\text{Zero}}{\text{Zero}}$$

$$\lim_{x \rightarrow 0} \frac{(x+2)^2 - 4}{x^2 + x} = \lim_{x \rightarrow 0} \frac{x^2 + 4x + 4 - 4}{x(x+1)} = \lim_{x \rightarrow 0} \frac{x(x+4)}{x(x+1)} = \lim_{x \rightarrow 0} \frac{(x+4)}{(x+1)} = 4$$

Example 8:

Find: $\lim_{x \rightarrow -5} \frac{x^3 + 125}{x^2 - 25}$

Solution

$$f(-5) = \frac{-125 + 125}{25 - 25} = \frac{\text{Zero}}{\text{Zero}}$$

$$\lim_{x \rightarrow -5} \frac{x^3 + 125}{x^2 - 25} = \lim_{x \rightarrow -5} \frac{(x+5)(x^2 - 5x + 25)}{(x-5)(x+5)}$$

$$= \lim_{x \rightarrow -5} \frac{(x^2 - 5x + 25)}{(x-5)} = \frac{-15}{2}$$

Exercise 1

Find the following limits:

(1) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

(2) $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 4}$

(3) $\lim_{x \rightarrow 0} \frac{x^2 - 2x + 3}{2x + 15}$

(4) $\lim_{x \rightarrow 1} \frac{3x^2 - x - 2}{x - 1}$

(5) $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + x}$

(6) $\lim_{x \rightarrow 1} \frac{2x^2 + x - 3}{x^2 - 1}$

(7) $\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x^2 - x - 2}$

(8) $\lim_{x \rightarrow -3} \frac{x^2 + 3x}{3x + 9}$

Lesson 4

Follow: Limit of the function algebraically(conjugate)

Example 1:

Find: $\lim_{x \rightarrow -1} \frac{\sqrt{5x+9}-2}{x+1}$

Solution

$$f(0) = \frac{2-2}{-1+1} = \frac{\text{Zero}}{\text{Zero}}$$

$$\lim_{x \rightarrow -1} \frac{\sqrt{5x+9}-2}{x+1} \times \frac{\sqrt{5x+9}+2}{\sqrt{5x+9}+2}$$

$$= \lim_{x \rightarrow -1} \frac{5x+9-4}{(x+1)(\sqrt{5x+9}+2)}$$

$$= \lim_{x \rightarrow -1} \frac{5(x+1)}{(x+1)(\sqrt{5x+9}+2)}$$

$$= \lim_{x \rightarrow -1} \frac{5}{(\sqrt{5x+9}+2)} = \frac{5}{4}$$

Note:

We can cancel the zero factor, by multiply each of the numerator and denominator by the conjugate of (the numerator, denominator or both).

Exercise for the student :

We can solve the following examples as previous example:

Example 2:

Find: $\lim_{x \rightarrow 5} \frac{\sqrt{x+4}-3}{x-5}$

Example 3:

Find: $\lim_{x \rightarrow 4} \frac{\sqrt{x-3}-1}{x-4}$

Exercise 2

Find the following limits:

(1) $\lim_{x \rightarrow 5} \frac{\sqrt{x-1}-2}{x-5}$

(2) $\lim_{x \rightarrow -1} \frac{x+1}{\sqrt{x+5}-2}$

(3) $\lim_{x \rightarrow 4} \frac{\sqrt{2x+1}-3}{x-4}$

(4) $\lim_{x \rightarrow -3} \frac{\sqrt{x+7}-2}{x+3}$

(5) $\lim_{x \rightarrow 0} \frac{x^2 + 2x}{\sqrt{x+9}-3}$

Lesson 5

Follow: Limit of the function algebraically

Theorem:

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n(a)^{n-1}$$

Corollary:

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m}(a)^{n-m}$$

Example 1:

Find: $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2}$

Solution

$$\lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2} = \lim_{x \rightarrow 2} \frac{x^5 - 2^5}{x - 2} = 5 \times 2^4 = 80$$

Example 2:

Find: $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$

Solution

$$\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2} = \lim_{x \rightarrow 2} \frac{x^4 - 2^4}{x - 2} = 4 \times 2^3 = 32$$

Example 3:

Find: $\lim_{x \rightarrow -1} \frac{x^5 + 1}{x^2 - 1}$

Solution

$$\lim_{x \rightarrow -1} \frac{x^5 + 1}{x^2 - 1} = \lim_{x \rightarrow -1} \frac{x^5 - (-1)^5}{x^2 - (-1)^2} = \frac{5}{2} \times (-1)^3 = \frac{-5}{2}$$

Example 3:

Find: $\lim_{x \rightarrow 0} \frac{(x+1)^3 - 1}{x}$

Solution

$$\lim_{x \rightarrow 0} \frac{(x+1)^3 - 1}{x} = \lim_{x+1 \rightarrow 1} \frac{(x+1)^3 - 1^3}{(x+1) - 1}$$

$$= 3 \times (1)^2 = 3$$

Exercise 3

Find the following limits:

(1) $\lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1}$

(2) $\lim_{x \rightarrow -3} \frac{x^4 - 81}{x^5 + 243}$

(3) $\lim_{x \rightarrow 3} \frac{x^5 - 243}{x^3 - 27}$

(4) $\lim_{x \rightarrow -2} \frac{x + 2}{x^5 + 32}$

(5) $\lim_{x \rightarrow 9} \frac{x - 9}{x^2 - 81}$

Lesson 6

Limit of the function at infinity

Theorem:

$$\lim_{x \rightarrow \infty} \frac{1}{x} = \text{zero}$$

Corollary:

$$\lim_{x \rightarrow \infty} \frac{a}{x} = \text{zero, where } a \in \mathbb{R}$$

$$\lim_{x \rightarrow \infty} \frac{a}{x^n} = \text{zero, where } a \in \mathbb{R} \text{ and } n \in \mathbb{R}^+$$

Note:

To find the limit of the function at infinity, we divide each of numerator and denominator by x raised to the higher power in denominator

Example 1:

Find: $\lim_{x \rightarrow \infty} \frac{3x^4 + x^3 - 2}{2x^4 + 3x^2 + 1}$

Solution

$$\lim_{x \rightarrow \infty} \frac{3x^4 + x^3 - 2}{2x^4 + 3x^2 + 1}$$

(Dividing both numerator and denominator by x^4)

$$= \lim_{x \rightarrow \infty} \frac{3 + \frac{1}{x} - \frac{2}{x^4}}{2 + \frac{3}{x^2} + \frac{1}{x^4}}$$

$$= \frac{3+0-0}{2+0+0} = \frac{3}{2}$$

Note:

If the higher power in numerator is equal to the higher power in denominator. Then the limit is coefficients of x with the higher power in numerator and denominator.

Example 2:

Find: $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{8x^3 + 5x} + x}{3x - 2}$

Solution

(Dividing both numerator and denominator by x)

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt[3]{8 + \frac{5}{x^2}} + 1}{3 - \frac{2}{x}} &= \frac{\sqrt[3]{8 + \text{zero}} + 1}{3 - \text{zero}} \\ &= \frac{2 + 1}{3} = 1 \end{aligned}$$

Example 3:

Find: $\lim_{x \rightarrow \infty} \frac{x + 2}{\sqrt{9x^2 + 25}}$

Solution

(Dividing both numerator and denominator by x)

$$\lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x}}{\sqrt{9 + \frac{25}{x^2}}} = \frac{1 + \text{zero}}{\sqrt{9 + \text{zero}}} = \frac{1}{3}$$

Example 4:

Find: $\lim_{x \rightarrow \infty} \frac{2x^4 - 3x^2 + 2}{4x^3 - 7x^2 + 5}$

Solution

(Dividing both numerator and denominator by x^3)

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2x - \frac{3}{x} + \frac{2}{x^3}}{4 - \frac{7}{x} + \frac{5}{x^3}} \\ = \frac{\infty - \text{zero} + \text{zero}}{4 - \text{zero} + \text{zero}} = \infty \end{aligned}$$

Note:

If the higher power in numerator is greater than the higher power in denominator. Then the limit is ∞

Example 5:

Find: $\lim_{x \rightarrow \infty} \frac{2x^3 + 11}{x^5 + 17}$

Solution

(Dividing both numerator and denominator by x^5)

$$\lim_{x \rightarrow \infty} \frac{\frac{2}{x^2} + \frac{11}{x^5}}{1 + \frac{17}{x^5}}$$

$$= \frac{\text{zero} + \text{zero}}{1 + \text{zero}} = \text{zero}$$

Note:

If the higher power in numerator is less than the higher power in denominator. Then the limit is Zero

Exercise 4

Find the following limits:

(1) $\lim_{x \rightarrow \infty} \frac{3x-1}{2-x}$

(2) $\lim_{x \rightarrow \infty} \frac{x+2}{\sqrt{9x^2+15}}$

(3) $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{8x^3+3x}-1}{5x-2}$

(4) $\lim_{x \rightarrow \infty} \frac{(x^2-3)(2x^3+12)}{x^5+8}$

(5) $\lim_{x \rightarrow \infty} \frac{(2x+3)^2}{x^2-x+1}$

(6) $\lim_{x \rightarrow \infty} \frac{2x^3-3x^2+2}{4x^2-3x^3+x}$

(7) $\lim_{x \rightarrow \infty} x^2 + 5x - x$

(8) $\lim_{x \rightarrow \infty} \frac{5x^3+x^2+1}{x^2+x}$

Lesson 7

Limit of the trigonometrical functions

Theorem:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

where x is in radian measure

Corollaries:

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin ax}{bx} = \frac{a}{b}$$

$$\lim_{x \rightarrow 0} \frac{\tan ax}{bx} = \frac{a}{b}$$

$$\lim_{x \rightarrow 0} \frac{\tan ax}{\sin bx} = \frac{a}{b}$$

Notice that:

$$\lim_{x \rightarrow 0} \frac{\sin^2 ax}{bx^2} = \frac{a^2}{b}$$

$$\lim_{x \rightarrow 0} \frac{\sin ax^2}{bx^2} = \frac{a}{b}$$

Corollary:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \text{zero}$$

Example (1): Find the following limits:

$$(1) \lim_{x \rightarrow 0} \frac{\sin 5x}{3x} = \frac{5}{3}$$

$$(2) \lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 5x} = \frac{3}{5}$$

$$(3) \lim_{x \rightarrow 0} \frac{\sin^2 2x}{3x^2} = \frac{4}{3}$$

Example (2):

Find: $\lim_{x \rightarrow 0} \frac{3x + \tan 5x}{2x + \sin x}$

Solution

(Dividing both numerator and denominator by x)

$$\lim_{x \rightarrow 0} \frac{3 + \frac{\tan 5x}{x}}{2 + \frac{\sin x}{x}} = \frac{3+5}{2+1} = \frac{8}{3}$$

Example (3):

Find: $\lim_{x \rightarrow 0} \frac{2x + \sin 3x}{5x + x \cos 5x}$

Solution

(Dividing both numerator and denominator by x)

$$\lim_{x \rightarrow 0} \frac{2 + \frac{\sin 3x}{x}}{5 + \cos 5x} = \frac{2+3}{5+1} = \frac{5}{6}$$

Example (4):

Find: $\lim_{x \rightarrow 0} \frac{3x^2 + \tan^2 5x}{5x \sin 3x}$

Solution

(Dividing both numerator and denominator by x^2)

$$\lim_{x \rightarrow 0} \frac{3 + \frac{\tan^2 5x}{x^2}}{\frac{5 \sin 3x}{x}} = \frac{3+25}{5 \times 3} = \frac{28}{15}$$

Example (5):

Find: $\lim_{x \rightarrow 0} \frac{2x \sin 3x}{\sin^2 5x - 5 \tan x^2}$

Solution

(Dividing both numerator and denominator by x^2)

$$\lim_{x \rightarrow 0} \frac{\frac{2 \sin 3x}{x}}{\frac{\sin^2 5x}{x^2} - \frac{5 \tan x^2}{x^2}} = \frac{2 \times 3}{25 - 5} = \frac{3}{10}$$

Example (6):

Find: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x}$

Solution

$$\lim_{x \rightarrow 0} \frac{\frac{1 - \cos x}{x}}{\frac{\sin x}{x}} = \frac{\text{zero}}{1} = \text{Zero}$$

Exercise 5

Find the following limits:

$$(1) \lim_{x \rightarrow 0} \frac{\sin 2x}{\tan 3x}$$

$$(2) \lim_{x \rightarrow 0} \frac{3x}{\sin x}$$

$$(3) \lim_{x \rightarrow 0} \frac{2x^2 + \tan 6x}{3x \cos 2x}$$

$$(4) \lim_{x \rightarrow 0} \frac{2x + \tan 6x}{3x \cos 2x}$$

$$(5) \lim_{x \rightarrow 0} \frac{5x - \sin 2x}{7x + \sin 2x}$$

$$(6) \lim_{x \rightarrow 0} \frac{\tan^2 2x}{\sin 3x^2}$$

$$(7) \lim_{x \rightarrow 0} \frac{\sin^2 3x}{\tan^2 2x}$$

$$(8) \lim_{x \rightarrow 0} \frac{x - x \cos x}{\sin^2 5x}$$

$$(9) \lim_{x \rightarrow 0} \frac{3x \sin 4x}{\tan 4x^2}$$

Lesson 8

The variation function

The variation function

If y is a function in the variable x . and x changes from x_1 to x_2 then the change in y is called as " variation function" and denoted by $V(h)$.

$$V(h) = f(x_2) - f(x_1)$$

If the change in x is denoted by (h) : $h = x_2 - x_1$

$$\therefore x_2 = x_1 + h$$

$$\therefore V(h) = f(x_1 + h) - f(x_1)$$

Example 1:

If $f(x) = x^2 + x - 3$, then find the variation function at $x = 2$

Solution

$$\begin{aligned} V(h) &= f(2 + h) - f(2) \\ &= (2 + h)^2 + (2 + h) - 3 - (4 + 2 - 3) \\ &= 4 + 4h + h^2 + 2 + h - 3 - 3 \\ &= h^2 + 5h \end{aligned}$$

Example 2:

Find the variation function of the function f where $f(x) = x^2 + 2$ when x changes from 3 to 3.1

Solution

$$\begin{aligned} V(h) &= f(3.1) - f(3) \\ &= (3.1)^2 + 2 - (9 + 2) \\ &= 0.61 \end{aligned}$$

Example 3:

If $f(x) = x^2 - x + 1$. Find the variation function of the function when $x = 3$ then find $V(0.2)$

Solution

$$\begin{aligned} V(h) &= f(3 + h) - f(3) \\ &= (3 + h)^2 - (3 + h) + 1 - (9 - 3 + 1) \\ &= 9 + 6h + h^2 - 3 - h + 1 - 7 \\ &= h^2 + 5h \end{aligned}$$

$$\therefore V(0.2) = (0.2)^2 + 5 \times 0.2 = 1.04$$

Exercise 6

- (1) If $f(x) = x^2 + 1$. Find the function of variation the when $x = 4$. then find $V(0.3)$
- (2) Find the function of variation the function f where $f(x) = x^2 - 2$ when x changes from 2 to 2.4
- (3) Find the function of variation of the function f where $f(x) = x^2$ when x changes from 5 to 5.2
- (4) If $f(x) = x^2 - 2x$. Find the function of variation of the function when $x = 3$. then find $V(0.1)$
- (5) If $f(x) = x^2 + 2x + 2$. Find the function of variation of the function when $x = 1$

Lesson 9

Function of the average rate of variation

The function of the average rate :

is denoted by $A(h)$, where

$$A(h) = \frac{V(h)}{h}$$

Example 1:

If $f(x) = x^2 + x - 3$, then find function of the average rate of change of the function at $x = 2$

Solution

$$\begin{aligned} V(h) &= f(2+h) - f(2) \\ &= (2+h)^2 + (2+h) - 3 - (4+2-3) \\ &= 4 + 4h + h^2 + 2 + h - 3 - 3 \\ &= h^2 + 5h \end{aligned}$$

$$\therefore A(h) = \frac{V(h)}{h} = \frac{h^2 + 5h}{h} = h + 5$$

Example 2:

Find the function of the average rate of change of the function f where $f(x) = x^2 + 2$ when x changes from 3 to 3.1

Solution

$$\begin{aligned} V(h) &= f(3.1) - f(3) \\ &= (3.1)^2 + 2 - (9 + 2) \\ &= 0.61 \end{aligned}$$

$$\therefore A(h) = \frac{V(h)}{h} = \frac{0.61}{3.1-3} = 6.1$$

Example 3:

If $f(x) = x^2 - x + 1$. Find the function average rate of change of the function when $x = 3$ then find $A(0.6)$

Solution

$$\begin{aligned} V(h) &= f(3 + h) - f(3) \\ &= (3 + h)^2 - (3 + h) + 1 - (9 - 3 + 1) \\ &= 9 + 6h + h^2 - 3 - h + 1 - 7 \\ &= h^2 + 5h \end{aligned}$$

$$\therefore A(h) = \frac{V(h)}{h} = \frac{h^2 + 5h}{h} = h + 5$$

$$\therefore A(h) = 0.6 + 5 = 5.6$$

Exercise 7

- (1) If $f(x) = x^2 + 2$. Find the function of the average rate of change of the function when $x = 4$. then find A (0.3)
- (2) Find the function of the average rate of change of the function f.
where $f(x) = x^2 - 1$. when x changes from 2 to 2.4
- (3) Find the function of the average rate of change of the function f.
where $f(x) = x^2$. when x changes from 3 to 3.2
- (4) If $f(x) = x^2 - 2x$. Find the function of the average rate of change of the function when $x = 3$. then find A (0.1)
- (5) If $f(x) = x^2 + 2x + 2$. Find the function of the average rate of change of the function when $x = 1$.

Lesson 10

The rate of change

The rate of change:

$$\text{The rate of change} = \lim_{h \rightarrow 0} A(h)$$

Example 1:

If $f(x) = x^2 + x - 3$, then find the rate of change of the function at $x = 2$

Solution

$$\begin{aligned} V(h) &= f(2 + h) - f(2) \\ &= (2 + h)^2 + (2 + h) - 3 - (4 + 2 - 3) \\ &= 4 + 4h + h^2 + 2 + h - 3 - 3 \\ &= h^2 + 5h \end{aligned}$$

$$\therefore A(h) = \frac{V(h)}{h} = \frac{h^2 + 5h}{h} = h + 5$$

$$\therefore \text{The rate of change} = \lim_{h \rightarrow 0} A(h) = \lim_{h \rightarrow 0} (h + 5) = 5$$

Example 2:

Find the rate of change of the function f where $f(x) = x^2 + 2$ when $x = 3$

Solution

$$\begin{aligned} V(h) &= f(h + 3) - f(3) \\ &= (h + 3)^2 + 2 - (9 + 2) \\ &= 9 + 6h + h^2 + 2 - 11 \\ &= h^2 + 6h \end{aligned}$$

$$\therefore A(h) = \frac{V(h)}{h} = \frac{h^2 + 6h}{h} = h + 6$$

$$\therefore \text{The rate of change} = \lim_{h \rightarrow 0} A(h) = \lim_{h \rightarrow 0} (h + 6) = 6$$

Example 3:

If $f(x) = x^2 - x + 1$. Find the rate of change of the function when $x = 3$.

Solution

$$\begin{aligned} V(h) &= f(3 + h) - f(3) \\ &= (3 + h)^2 - (3 + h) + 1 - (9 - 3 + 1) \\ &= 9 + 6h + h^2 - 3 - h + 1 - 7 \\ &= h^2 + 5h \end{aligned}$$

$$\therefore A(h) = \frac{V(h)}{h} = \frac{h^2 + 5h}{h} = h + 5$$

$$\therefore \text{The rate of change} = \lim_{h \rightarrow 0} A(h) = \lim_{h \rightarrow 0} (h + 5) = 5$$

Exercise 8

- (1) If $f(x) = x^2 + 2$. Find the rate of change of the function when $x = 4$.
- (2) Find the rate of change of the function f . where $f(x) = x^2 - 1$. when $x = 2$
- (3) Find the average rate of change function of the function f . where $f(x) = x^2$. when $x = 3$
- (4) If $f(x) = x^2 - 2x$. Find the rate of change of the function f . when $x = 3$.
- (5) If $f(x) = x^2 + 2x + 2$. Find the rate of change of the function f . when $x = 1$.

Unit Test

First Question: Find the following limits:

(1) $\lim_{x \rightarrow 0} \frac{x^2 - 2x + 3}{2x + 15}$

(2) $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + x}$

Second Question:

(1) $\lim_{x \rightarrow 5} \frac{\sqrt{x-1} - 2}{x - 5}$

(2) Find the rate of change of the function f . where $f(x) = x^2 - 5$. when $x = 2$

Third Question:

(1) $\lim_{x \rightarrow 3} \frac{x^5 - 243}{x^3 - 27}$

(2) If $f(x) = x^2 - 2x$. Find the function of the average rate of change of the function when $x = 3$. then find $A(0.1)$

Fourth Question:

(1) $\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan 3x}$

(2) $\lim_{x \rightarrow 0} \frac{5x - \sin 2x}{7x + \sin 2x}$

Evaluation

Applied Technology Schools

Second Unit Statistics

Dear student, by the end studying of this unit you should have the following abilities and knowledges:

- ❖ To find the mean.
- ❖ To find the median.
- ❖ To find the mode.
- ❖ To find the standard deviation.
- ❖ To know the concept of the random experiment.
- ❖ To know the concept of the sample space.
- ❖ To write the sample space for some random experiments.
- ❖ To know the concepts of (the event, the prime event, the certain event and impossible event)
- ❖ To know the mutually exclusive events.
- ❖ To know the concepts of the operations on the events.
- ❖ To use the axioms of probability to calculate occurrence event.

Lesson 1

The Arithmetic Mean

❖ The arithmetic mean of set of values = $\frac{\text{sum of the values}}{\text{number of the values}}$

Example 1:

Find the arithmetic mean of the following set: 7, 3, 5 and 1

Solution

$$\text{The arithmetic mean} = \frac{7+3+5+1}{4} = 4$$

Example 2:

If the arithmetic mean of values: 6, 5, 8, x, 4 is 6. Then find the value of x

Solution

$$\text{The arithmetic mean} = \frac{6+5+8+x+4}{5} = 6$$

$$23 + x = 30$$

$$\therefore x = 7$$

❖ The arithmetic mean of a frequency distribution:

Example 3:

Find the arithmetic mean of the following frequency distribution:

Sets	2 -	4 -	6 -	8 -	10 -	Total
frequency	4	6	4	3	3	20

Solution

Sets	Centre of set (x)	Frequency (k)	$x \times k$
2 -	3	4	12
4 -	5	6	30
6 -	7	4	28
8 -	9	3	27
10 -	11	3	33
Sum		20	130

The arithmetic mean = $\frac{\Sigma (x \times k)}{\Sigma k} = \frac{130}{20} = 6.5$

To find the center of the first set we add the first two sets then divide the sum by 2. And so

Exercise 1

- (1) Find the arithmetic mean of the set of values: 4, 8, 3, 2 and 9
- (2) If the arithmetic mean of values: 10, 5, x, 6, 2 is 4. Then find the value of x
- (3) Find the arithmetic mean of the following frequency distribution:

Sets	0 -	4 -	8 -	12 -	16 -	Total
frequency	1	3	5	4	7	20

- (4) Find the arithmetic mean of the following frequency distribution:

Sets	5 -	15 -	25 -	35 -	45 -	55 -	Total
frequency	4	16	18	25	18	19	100

Lesson 2

The Median

❖ The median of set values is the value which lies in the middle of those values.

(1) If the number of values is an odd number:

Example 1:

Find the median of the following values: 18, 25, 11, 9 and 15

Solution

Arrangement of values:

9, 11, 15, 18 and 25

The median = 15

Example 2:

Find the median of the following values: 10, 12, 8, 6, 15, 19 and 7

Solution

Arrangement of values:

6, 7, 8, 10, 12, 15 and 19

The median = 10

(2) If the number of values is an even number:

Example 1:

Find the median of the following values: 4, 10, 6, 12, 7, 9, 5 and 14

Solution

Arrangement of values:

4, 5, 6, 7, 9, 10, 12 and 14

(3) The median = $\frac{7 + 9}{2} = 8$

❖ The median of frequency distribution:

Example 3:

Find the median of the following frequency distribution:

Sets	2 -	4 -	6 -	8 -	10 -	Total
frequency	1	9	15	20	5	50

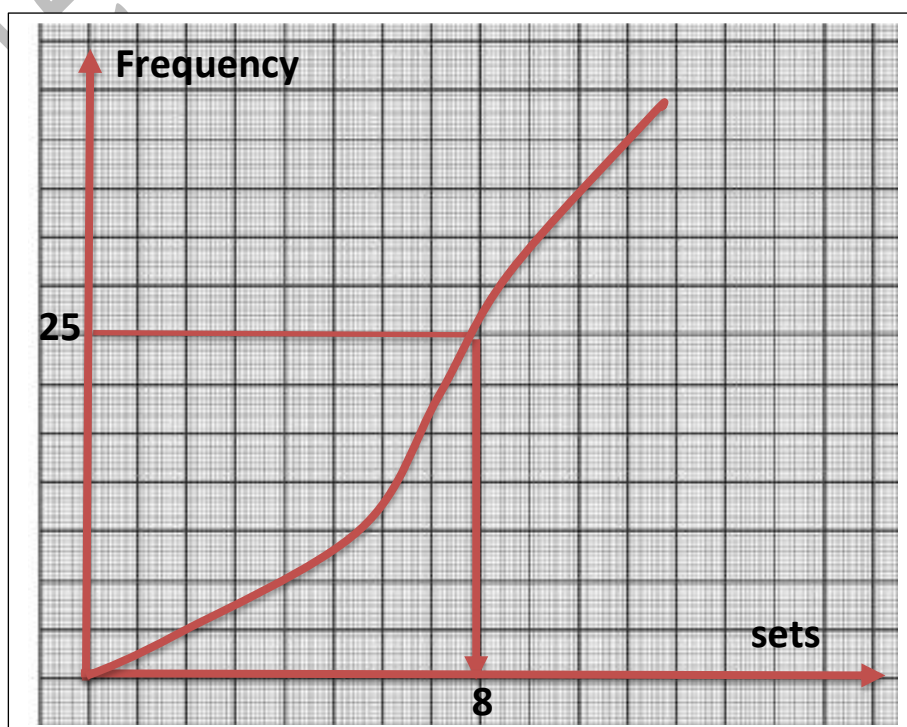
Solution

- We form the ascending cumulative frequency table

Upper limits of sets	Frequency
Less than 2	0
Less than 4	1
Less than 6	10
Less than 8	25
Less than 10	45
Less than 12	50

$$\text{Order of median} = \frac{50}{2} = 25$$

The median = 8



Exercise 2

- (1) Find the median of the set of values: 3, 5, 8, 7 and 4
- (2) Find the median of the set of values: 2, 5, 1, 7, 3 and 6
- (3) Find the median of the following frequency distribution:

Sets	0 -	4 -	8 -	12 -	16 -	Total
frequency	5	15	30	40	10	100

- (4) Find the median of the following frequency distribution:

Sets	0 -	5 -	10 -	15 -	20 -	25 -	30 -	35 -	Total
frequency	0	1	5	8	12	16	6	2	50

Lesson 3

The Mode

❖ The mode of set of values is the most common value.

Example 1:

Find the mode of the following vales:

5, 6, 7, 6 and 9

Solution

The mode = 6

Example 2:

Find the mode of the following vales:

11, 12, 11, 13, 12, 14, 12 and 12

Solution

The mode = 12

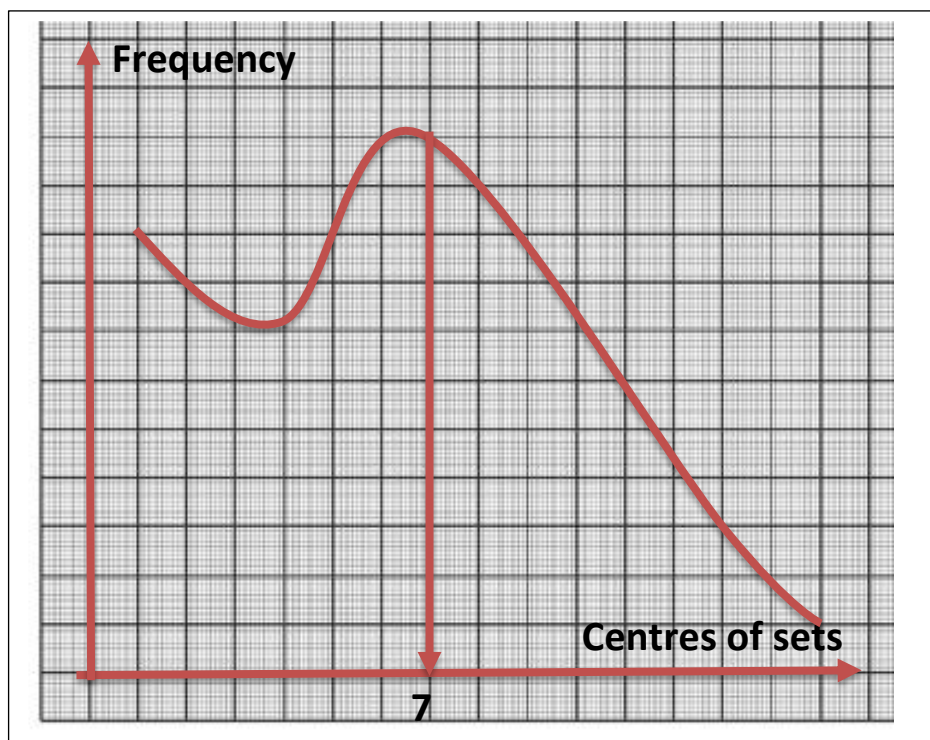
❖ The median of frequency distribution:

Find the median of the following frequency distribution:

Sets	2 -	4 -	6 -	8 -	10 -	Total
frequency	1	9	15	20	5	50

Solution

We draw the frequency curve of this distribution



- The height of the top of the curve on horizontal axis is the mode
The mode = 7

• Note

We can use the histogram to find the mode.

Exercise 3

(1) Find the mode of the set of values: 3, 2, 5, 2, 7 and 2

(2) Find the mode of the following frequency distribution:

Sets	0 -	4 -	8 -	12 -	16 -	Total
frequency	5	15	30	40	10	100

(4) Find the mode of the following frequency distribution:

Sets	0 -	5 -	10 -	15 -	20 -	25 -	30 -	35 -	Total
frequency	0	1	5	8	12	16	6	2	50

Lesson 4

The standard deviation

❖ The standard deviation of set of values:

Example 1:

Calculate the standard deviation of the values:

12, 13, 16, 18 and 21

Solution

The arithmetic mean of the values $(\bar{x}) = \frac{80}{5} = 16$

x	$x - \bar{x}$	$(x - \bar{x})^2$
12	- 4	16
13	-3	9
16	0	0
18	2	4
21	5	25
80		54

$$\begin{aligned}
 \text{The standard deviation } (\delta) &= \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}} \\
 &= \sqrt{\frac{54}{5}} = 3.286
 \end{aligned}$$

❖ The standard deviation of frequency distribution:**Example 2:**

The following frequency distribution shows the number of damaged units which found in 100 boxes:

No. of damaged units	0	1	2	3	4	5
No. of boxes	3	16	17	25	20	19

Find the standard deviation of the damaged units.

Solution

x	k	X × k	x - $\bar{x}$	(x - $\bar{x}$) ²	(x - $\bar{x}$) ² × k
0	3	0	-3	9	27
1	16	16	-2	4	64
2	17	34	-1	1	17
3	25	75	0	0	0
4	20	80	1	1	20
5	19	95	2	4	76
Sum	100	300			204

The arithmetic mean ($\bar{x}$) = $\frac{300}{100} = 3$

$$\begin{aligned} \text{The standard deviation (} \delta \text{)} &= \sqrt{\frac{\Sigma (x - \bar{x})^2 \times k}{\Sigma k}} \\ &= \sqrt{\frac{204}{100}} = 1.428 \end{aligned}$$

Exercise 4

- (1) Find the standard deviation of the set of values: 8, 9, 7, 6 and 5
- (2) Find the standard deviation of the set of values: 25, 24, 25, 30, 28 and 30
- (3) The following frequency distribution shows the number of children in 100 families:

No. of children	0	1	2	3	4	Total
No. of children	8	16	50	20	6	100

Calculate the standard deviation.

- (4) Find the standard deviation of the following frequency distribution:

No. of damaged units	0	3	5	7	9	10	Total
No. of boxes	4	17	15	30	20	14	100

Lesson 5

Application on central tendency and

Answer the following questions:

(Student should solve the exercises by himself)

(1) If the arithmetic mean of values: 6, 5, 8, x , 4 is 6. Then find the value of x

Solution

(2) Find the median of the values: 4, 10, 6, 12, 7, 9, 5 and 14

Solution

(3) Find the mode of the values: 11, 12, 11, 13, 12, 14 and 12

Solution

- (4) Calculate the standard deviation of the set of values: 12, 13, 16, 18 and 21
Solution

- (5) Find the arithmetic mean of the following frequency distribution:

Sets	0 -	4 -	8 -	12 -	16 -	Total
frequency	1	3	5	4	7	20

Solution

(6) Find the median of the following frequency distribution:

Sets	2 -	4 -	6 -	8 -	10 -	Total
frequency	1	9	15	20	5	50

Solution

(7) Find the mode of the following frequency distribution:

Sets	2 -	4 -	6 -	8 -	10 -	Total
frequency	3	10	12	10	5	40

Solution

(8) The following frequency distribution shows the number of damaged units which found in 100 boxes:

No. of defective units	0	1	2	3	4	5
No. of boxes	3	16	17	25	20	19

Find the standard deviation of the damaged units.

Solution

Lesson 6

The Probability (key-terms and concepts)

The random experiment:

It is an experiment in which we can specify all its possible outcomes before performing it, but we cannot determine which outcome will occur certainly.

The sample space:

It is the set of all possible outcomes of a random experiment.

Example 1:

Write the sample space of throwing a coin once and observing the apparent face.

Solution

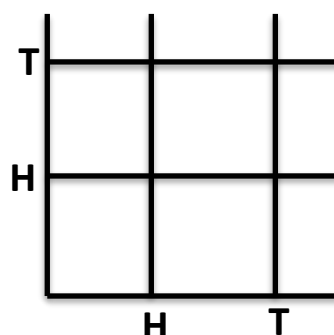
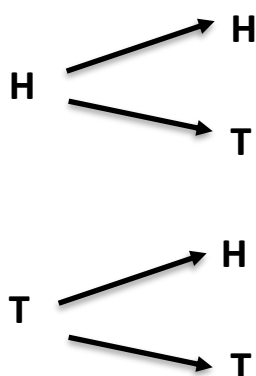
$$S = \{H, T\}$$

Example 2:

Write the sample space of throwing a coin twice and observing the apparent face.

Solution

$$S = \{(H,H), (H,T), (T,H), (T,T)\}$$



Example 3:

A bag contains three identical balls of except the color one of them is red, the second is black and the third is white. Write the sample space of the following:

- (a) Drawing a ball randomly and observing its color.
- (b) Drawing two balls one after the other with observing the color.

Solution

(a) $S = \{ r, b, w \}$

(b) $S = \{ (r,r), (r,b), (r,w), (b,r), (b,b), (b,w), (w,r), (w,b), (w,w) \}$

The event:

It is a subset of the sample space.

Types of the events:

- (1) The prime (simple) event: it content one element.
- (2) Certain event: it is an event preceded to occur.
- (3) Impossible event: it is an event impossible to occur.
- (4) Two mutually exclusive events: two events, if one of them occur the other not occur

Notice that:

- (1) The prime number: a number has two factors are whole one and itself.
for example: 2, 3, 5, 7, 11, 13, 17, 19,
- (2) The even number: any number which is divisible by 2.
- (3) The odd number: any number which is not divisible by 2.

Examples:

As throwing a die twice and observing the apparent face.

Write down the following events:

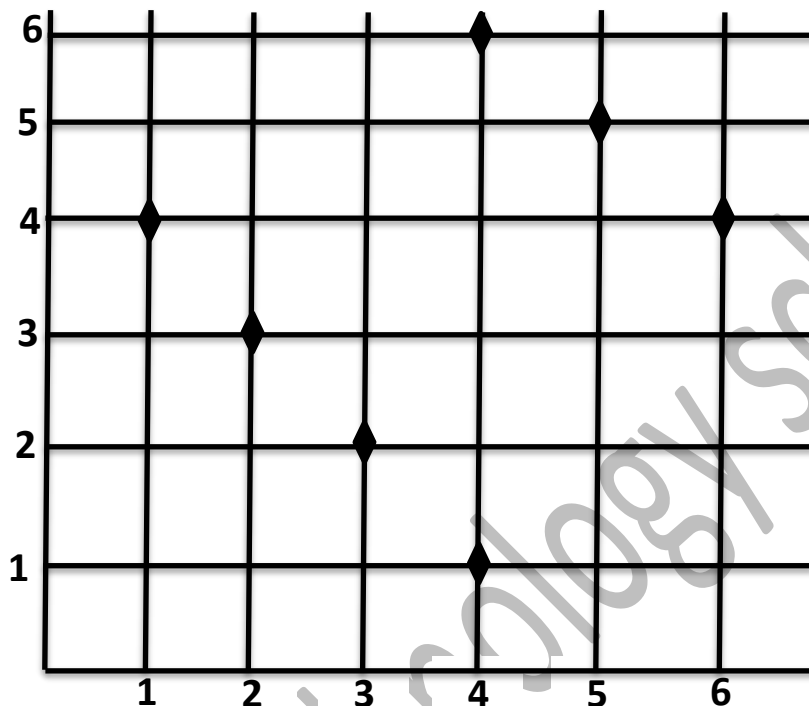
- (1) The event of apparent two numbers their sum is 9

(2) The event of apparent two numbers their sum is divisible by 5

Solution

(1) $A = \{(3,6), (4,5), (5,4), (6,3)\}$

(2)



$B = \{(1,4), (2,3), (3,2), (4,1), (4,6), (5,5), (6,4)\}$

Exercise 5

(1) A box contains three identical balls numbered from 1 to 3, if two balls are drawn at random one after the other with replacement and observing the apparent number of drawn ball. Write the sample space.

(2) In experiment of throwing a coin then a die and observing their upper face. Write the sample space of this experiment then determine the following events:

- The event of apparent head and an odd number.
- The event of apparent tail and an even number.
- The event of apparent a prime number greater than 2.
- The event of apparent a number divisible by 3.

(3) In experiment of throwing a die twice and observing the upper face. determine the following events:

- The event of apparent two equal numbers.
- The event of apparent two numbers their sum is 9.
- The event of apparent two numbers their sum is 13.
- The event of apparent number is 3 once at least.

Lesson 7

The Axioms of Probability

(1) For all $A \subset S$, there is a real number $P(A)$ is called probability of the event A

where:

$$0 \leq P(A) \leq 1$$

$$(2) P(S) = 1$$

(3) If $A \subset S$, $B \subset S$ and the events A, B are mutually exclusive events then:

$$P(A \cap B) = 0$$

Probability Calculate:

If "S" is sample space of a random experiment then probability of occurring

of event $A \subset S$ is:

$$P(A) = \frac{m}{n}$$

where (m) number of elements of A,
(n) number of elements of S.

Example 1:

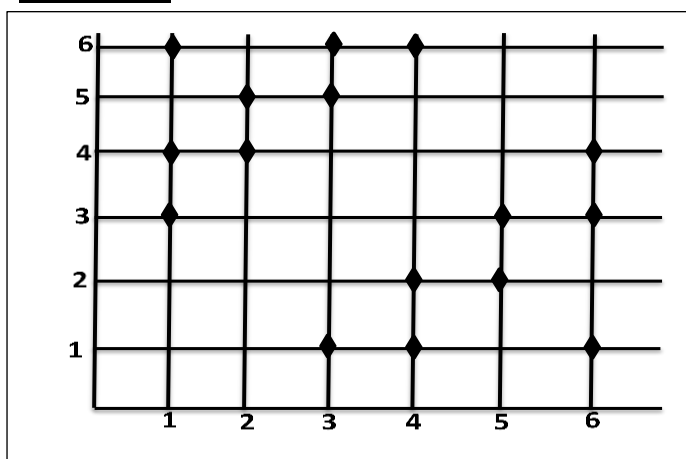
As throwing a die twice and observing the apparent number on upper face.
Find:

- (a) The probability of sum of the two numbers is divisible by 6.
- (b) The probability of absolute difference of the two numbers equals a prime number.

Solution

$$P(A) = \frac{6}{36} = \frac{1}{6}$$

$$P(A) = \frac{16}{36} = \frac{4}{9}$$



Example 2:

A bag contains 30 identical cards numbered from 1 to 30, a card is drawn randomly. Find the probability of written number on the card is

- (1) An even number and divisible by 5
- (2) Divisible by 3 or 5.

Solution

$$A = \{ 10, 20, 30 \}$$

$$P(A) = \frac{3}{30} = 0.1$$

$$B = \{ 3, 5, 6, 9, 10, 12, 15, 18, 20, 21, 24, 25, 27, 30 \}$$

$$P(B) = \frac{14}{30} = \frac{7}{15}$$

Exercise 6

(1) From the set of the digits: 2, 3, 4, 5 and 6. Form a 2-digit different number then find:

* Probability of the number whose tens digit is odd.

* Probability of the number whose tens digit is odd, or units digit is prime.

(2) A box contains identical and colored balls, 4 of them are red, 6 are blue and 5 are yellow. If a ball is draw randomly.

Calculate the probability of the ball is:

- red.
- Blue or yellow.
- Not blue.
- Neither red nor yellow.

(3) A set of identical cards and numbered from 1 to 30, a card is drawn randomly. Find the probability of written number on the card is

- (1) A number divisible by 3.
- (2) A number divisible by 5.
- (3) A number divisible by 3 and 5.
- (4) A number divisible by 3 or 5.

Lesson 8

The Language of Probability

Administration and Operation Unit of Applied Technology Schools.

Symbol	The meaning
$P(A \cap B)$	Probability of occurring A and B together.
$P(A \cup B)$	Probability of occurring A or B.
	Probability of occurring one of them at least.
	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$P(A^c)$	Probability of non-occurrence A
	$P(A^c) = 1 - P(A)$
$P(A - B)$	Probability of occurrence A only.
	Probability of occurrence A and non- occurrence B.
	$P(A - B) = P(A \cap B^c)$
	$P(A - B) = P(A) - P(A \cap B)$
$P(A^c \cap B^c)$	Probability of non-occurrence any of A or B.
	$P(A^c \cap B^c) = 1 - P(A \cup B)$
$P(A^c \cup B^c)$	Probability of non-occurrence any of A and B together.
	Probability of occurrence one of at most.
	$P(A^c \cup B^c) = 1 - P(A \cap B)$
Probability of occurrence A or B only.	Probability of occurrence one of them only.
	$P(A \cup B) - P(A \cap B)$

Example 1:

If A and B are two events of the sample space S, $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{8}$, $P(A \cap B) = \frac{1}{4}$. Find: $P(A^c)$, $P(B^c)$, $P(A \cup B)$ and $P(A^c \cap B^c)$

Solution

$$P(A^c) = 1 - P(A) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(B^c) = 1 - P(B) = 1 - \frac{3}{8} = \frac{5}{8}$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{1}{3} + \frac{3}{8} - \frac{1}{4} = \frac{11}{24} \end{aligned}$$

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - \frac{11}{24} = \frac{13}{24}$$

Example 2:

Two soldiers shoot at target at the same time, if the probability that first soldier hits the target is $\frac{1}{2}$, the probability that second soldier hits the target is $\frac{2}{3}$ and the probability that the two soldiers hit the target together is $\frac{1}{3}$.

Find:

- The probability of hitting the target.
- The probability of hitting the target by first soldier only.

Solution

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{2}{3}$$

$$P(A \cap B) = \frac{1}{3}$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{1}{2} + \frac{2}{3} - \frac{1}{3} = \frac{5}{6} \end{aligned}$$

$$\begin{aligned} P(A - B) &= P(A) - P(A \cap B) \\ &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \end{aligned}$$

Example 3:

If X and Y are two events of the sample space, $P(X) = \frac{5}{8}$, $P(Y^c) = \frac{2}{5}$, $P(X \cap Y) = \frac{2}{5}$. Find:

- Probability of occurrence one of the two events at most.
- Probability of occurrence one of two events only.

Solution

$$P(Y) = 1 - P(Y^c) = 1 - \frac{2}{5} = \frac{3}{5}$$

- $P(X \cup Y^c) = 1 - P(X \cap Y) = 1 - \frac{2}{5} = \frac{3}{5}$

$$\begin{aligned} P(X \cup Y) &= P(X) + P(Y) - P(X \cap Y) \\ &= \frac{5}{8} + \frac{3}{5} - \frac{2}{5} = \frac{33}{40} \end{aligned}$$

- Probability of occurrence one of them only
 $= P(X \cup Y) - P(X \cap Y)$
 $= \frac{33}{40} - \frac{2}{5} = \frac{17}{40}$

Exercise 7

Administration and Operation Unit of Applied Technology Schools.

(1) If A and B are two events of the sample space S, $P(A) = 0.3$, $P(B) = 0.5$, $P(A \cap B) = 0.12$.

Find: $P(A^c)$, $P(A \cup B)$ and $P(A^c \cap B^c)$.

(2) If A and B are two events of the sample space a random experiment, where $P(A) = 0.5$, $P(A \cup B) = 0.75$, $P(B^c) = 0.7$

Find: $P(A \cap B)$, $P(A^c \cup B^c)$ and $P(B \cap A^c)$

(3) If A and B are two events of the sample space a random experiment, where $P(A \cup B) = 0.6$, $P(A - B) = 0.25$

Find: $P(A)$ and $P(B)$.

(2) If A and B are two events of the sample space a random experiment, where $P(A) = 0.4$, $P(B^c) = 3P(B)$ and $P(A \cap B) = 0.2$ find:

- Probability of occurrence A only.
- Probability of occurring A or B.
- Probability of occurrence A and non-occurrence B.

Lesson 9

Follow: The Language of Probability

Example 1:

If A and B are two events of the sample space S, $P(A) = 0.3$, $P(B) = 0.5$, $P(A \cap B) = 0.12$

Find: $P(A^c)$, $P(A \cup B)$ and $P(A^c \cap B^c)$.

Solution

- $P(A^c) = 1 - P(A) = 1 - 0.3 = 0.7$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 0.3 + 0.5 - 0.12 = 0.68$
- $P(A^c \cap B^c) = 1 - P(A \cup B)$
 $= 1 - 0.68 = 0.32$

Example 2:

If A and B are two events of the sample space random experiment, $P(A) = \frac{3}{4}$, $P(B) = \frac{5}{8}$, $P(A^c \cap B^c) = \frac{1}{8}$, Find: $P(B^c)$ and $P(A \cap B)$.

Solution

- $P(B^c) = 1 - P(B) = 1 - \frac{5}{8} = \frac{3}{8}$
- $P(A^c \cap B^c) = 1 - P(A \cup B)$
 $\frac{1}{8} = 1 - P(A \cup B)$
 $\therefore P(A \cup B) = \frac{7}{8}$
- $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
 $= \frac{3}{4} + \frac{5}{8} - \frac{7}{8}$
 $= \frac{1}{2}$

Example 3:

If A and B are two mutually exclusive events of the sample space random experiment, $P(A - B) = \frac{1}{2}$, $P(A \cup B) = \frac{3}{5}$, Find: $P(A)$, $P(B)$ and $P(A' \cup B')$.

Solution

A and B are two mutually exclusive events

$$\therefore P(A \cap B) = \text{zero.}$$

- $P(A - B) = P(A) - P(A \cap B)$

$$\therefore P(A) = \frac{1}{2}$$

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\frac{3}{5} = \frac{1}{2} + P(B)$$

$$\therefore P(B) = \frac{1}{10}$$

- $P(A' \cup B') = 1 - P(A \cap B) = 1$

Exercise 8

(1) Three persons A, B and C are competing in a race, if the probability that B wins is equal to double of the probability that A wins, if the probability that C wins is equal to triple of the probability that A wins and one of them will be the winner. Find:

- The probability that A is not win.
- The probability that A or C wins.

(2) If $S = \{A, B, C\}$, $P(A) : P(A^c) = \frac{8}{3}$ and $P(B) : P(B^c) = \frac{5}{2}$ Find: $P(C) : P(C^c)$

(3) If A and B are two events of the sample space random experiment where: $P(B) = 3P(A)$, $P(A \cup B) = 0.75$.

Find $P(A)$ and $P(B)$ in the following cases:

(1) A and B are two mutually exclusive events.

(2) $A \subset B$

(4) If A and B are two mutually exclusive events, $P(A) = 0.4$, $P(B) = 0.3$

Find: $P(A \cup B)$

Answer the following questions:

(Student should solve the exercises by himself)

(1) If A and B are two events of the sample space of a random experiment,

$P(A) = \frac{1}{4}$, $P(B) = \frac{3}{8}$, $P(A \cap B) = \frac{1}{8}$. Find: $P(B^c)$ and $P(A^c \cap B^c)$.

Solution

(2) S is sample space of random experiment, all outcomes of it are equal in occurrence, A and B are two events from S. If $P(A \cup B) = \frac{5}{6}$, $P(B) = \frac{5}{12}$, the number of outcomes of event A equals 13 and number of all outcomes equals 24. Find

(a) Probability of occurring A and B together.

(b) $P(A^c \cup B^c)$

Solution

(3) If A and B are two events of the sample space random experiment,

$P(A) = 0.6$, $P(B) = 0.5$, $P(A \cup B) = 0.12$ find the probability of:

(a) occurring one of them at least. (b) occurrence A only.

Solution

(4) As throwing a die twice and observing the apparent number on upper face.

Find The probability of:

(a) sum of the two numbers is greater than or equals 9.

(b) one of the two numbers is 3 and their sum is less than 7.

Solution

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(5) A box contains 8 identical cards numbered from 1 to 8, two cards are drawn randomly one after the other with replacement. Find the probability of:

- **The absolute difference of the two numbers equals 3.**
- **Sum of the two numbers is less than 8.**

Solution

(6) A class contains 50 students. 25 of them are studying Chemistry, 20 of them are studying History and 15 students are studying Chemistry and History. A student is chosen randomly, find the probability that this student is studying Chemistry or History.

Solution

Answer the following questions:

First Question:

- (1) If A and B are mutually exclusive events, $P(A) = 0.4$, $P(B) = 0.3$ Find $P(A \cup B)$.
- (2) Find the mode of the set of values: 3, 2, 5, 2, 7 and 2.

Second Question:

- (1) Find the arithmetic mean of the following frequency distribution:

Sets	0 -	4 -	8 -	12 -	16 -	Total
frequency	1	3	5	4	7	20

- (2) From the set of the digits: 2, 3, 4, 5 and 6. Form a 2-digit different number then find Probability of the number whose tens digit is odd.

Third Question:

- (1) Find the median of the set of values: 2, 5, 1, 7, 3 and 6.
- (2) If A and B are two events of the sample space S, $P(A) = 0.3$, $P(B) = 0.5$, $P(A \cap B) = 0.12$
Find: $P(A^c)$, $P(A \cup B)$ and $P(A^c \cap B^c)$.

Fourth Question:

- (1) Find the standard deviation of the following frequency distribution:

No. of damaged units	0	3	5	7	9	10	Total
No. of boxes	4	17	15	30	20	14	100

- (2) A bag contains 30 identical cards numbered from 1 to 30, a card is drawn randomly. Find the probability of written number on the card is an even number and divisible by 5.

Evaluation

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